recursion.

A recursive function calls itself. Imagine a process would iterate indefinitely if not stopped by some condition! Such a process is known as infinite iteration. The condition that is applied in any recursive function is known as base condition. A base condition is must in every recursive function otherwise it will continue to execute like an infinite loop.

Overview of how recursive function works

1. Recursive function is called by some external code.
2. If the base condition is met then the program gives meaningful output and exits.
3. Otherwise, function does some required processing and then calls itself to continue recursion.

Here is an example of recursive function used to calculate factorial.

Example : 7.10

```
def fact(n):
        if n == 0:
                return 1
        else:
                return n * fact (n-1)
print (fact (0))
print (fact (5))
```

Output:

```
1
120
```

print(fact (2000)) will give Recursion Error after maximum recursion depth exceeded in comparison. This happens because python stops calling recursive function after 1000 calls by default. It also allows you to change the limit using sys.setrecursionlimit (limit_value).

Example:

```
import sys
sys.setrecursionlimit(3000)
def fact(n):
        if n == 0:
                return 1
         else:
                return n * fact(n-1)
print(fact (2000))
```


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